

# Solve missing-number expressions

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Solve  $6 \times (n + 5) = 72$ . Copy or complete the first step.

2. Solve  $180 \div (n - 2) = 30$ .

3. Find  $n$ :  $14 + n \times 5 = 64$ .

4. Miro says: Miro divides  $72$  by  $6$  but forgets to subtract the  $5$  inside the brackets. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Solve  $6 \times (n + 5) = 72$ . Copy or complete the first step.  
**Answer**  $n = 7$ .
2. Solve  $180 \div (n - 2) = 30$ .  
**Answer**  $n = 8$ .
3. Find  $n$ :  $14 + n \times 5 = 64$ .  
**Answer**  $n = 10$ .
4. Miro says: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets. Is this correct? Explain.  
**Answer** Undo the outer multiplication first, then undo the addition inside the brackets.

# Solve missing-number expressions

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Solve  $180 / (n - 2) = 30$ .

2. Find  $n$ :  $14 + n \times 5 = 64$ .

3. Create two different expressions using  $n = 4$  that both equal 36.

4. Identify and correct this misconception: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.

5. Write a complete sentence explaining how you checked one answer.

# Teacher answers and checking prompts

1. Solve  $180 / (n - 2) = 30$ .

**Answer**  $n = 8$ .

2. Find  $n$ :  $14 + n \times 5 = 64$ .

**Answer**  $n = 10$ .

3. Create two different expressions using  $n = 4$  that both equal 36.

**Answer** Examples include  $5 \times (4 + 3) + 1$  and  $40 - 4$ .

4. Identify and correct this misconception: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.

**Answer** Undo the outer multiplication first, then undo the addition inside the brackets.

5. Write a complete sentence explaining how you checked one answer.

**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

# Solve missing-number expressions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Create two different expressions using  $n = 4$  that both equal 36.

2. Explain why this reasoning fails: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.

3. Create a new example that tests the same idea as: Find  $n$ :  $14 + n \times 5 = 64$ .

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

# Teacher answers and checking prompts

1. Create two different expressions using  $n = 4$  that both equal 36.  
**Answer** Examples include  $5 \times (4 + 3) + 1$  and  $40 - 4$ .
2. Explain why this reasoning fails: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.  
**Answer** Undo the outer multiplication first, then undo the addition inside the brackets.
3. Create a new example that tests the same idea as: Find  $n$ :  $14 + n \times 5 = 64$ .  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.